Ab initio calculations on the magnetic properties of transition metal complexes

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Ab Initio Calculations on the Magnetic Properties of Transition Metal Complexes

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Abstract. We present a protocol for the ab initio determination of the magnetic properties of mono- and polynuclear transition metal compounds. First, we obtain the low lying electronic states by multireference methods. Then, we include spin-orbit coupling and an external magnetic field for the determination of zero-field splitting and g-tensors. For the polynuclear complexes the magnetic exchange coupling constants are determined by a modified complete active space self consistent field method. Based on the results of the ab initio calculations, magnetic data such as magnetic susceptibility or magnetization are simulated and compared to experimental data. The results obtained for the polynuclear complexes are further analysed by calculations on model complexes where part of the magnetic centers are substituted by diamagnetic ions. The methods are applied to different Co and Ni containing transition metal complexes.

Keywords: transition metals, quantum chemical calculations, magnetism, spin-orbit coupling
PACS: 31.15

INTRODUCTION

In material sience, there is a great interest in materials which can be manipulated externally. We have recently investigated a compound where the local Fe(II) center changes from low-spin to high-spin when the ligands of the complex are charged.[1] A change between low-spin and high-spin can also be induced by pressure or light. Another important class of such compounds are molecular magnets.[2, 3] In the latter, a special orintation of the spin of a molecule can be enforced by an external magnetic field. Here, an isolated molecule can show hysteresis in an magnetic field and solves as a model for storage of information on a molecular scale.

Our long term aim is the prediction of new molecules with interesting magnetic properties. Appropriate methods have to be developed and validated by comparison of the results of quantum chemical calculations of these properties to experimental data.

There are in general two approaches to calculate the magnetic exchange coupling in polynuclear transition metal centers, the broken symmetry approach within density functional theory, and the use of multireference wave functions. Here, we limit our considerations to the wave function approach which we applied to Co and Ni containing transition metal complexes.[5, 6] For the latter metal ions, there are quite often spatially almost degenerate states which make it necessary to include spin-orbit coupling in the calculations.

THEORETICAL METHODS

All geometry optimizations are performed with the program package TURBOMOLE[23]. The B3-LYP functional[21] was employed together with D3-BJ dispersion correction.[19, 20] The geometry was optimized for the broken symmetry solution of the Kohn-Sham equations[16, 17] using grid m4, def2-TZVP[22] basis for S, V, Co, Ni, and def2-SVP[22] basis for C,H.

All multireference calculations were performed with the Bochum suite of ab initio methods. The package includes a restricted open shell SCF program[9, 10], a complete active space SCF (CASSCF) program[11] a spin-orbit CI (SOCI) program[13] and a Multireference CEPA program[12]. In the SOCI program it is possible to include the Zeeman interaction with an external magnetic field. In the calculations, the metal ions were equipped using modified def2-SVP/def2-TZVP basis set ((13s9p5d→[4s3p3d] for Co^{2+}, and (14s9p7d)→[5s3p4d] for Ni^{2+}). For all other atoms, a def2-SVP basis set was used.

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Zero Field Splitting and G-factors of Mononuclear Metal Clusters

The calculations on zero-field splittings and g-factors is based on the geometric structures optimized by density functional theory.

For the given structures we obtain all necessary electronic states by CASSCF calculations. The five 3d-orbitals are included in the active space and optimized for the state average over all high-spin states. With these orbitals the energies and wave functions of all possible states within the 3d shell are calculated and considered in the following SOCI. The interaction of the electronic states with an external magnetic field is obtained by finite pertubation theory:

$$\hat{H}(B) = \hat{H}_0 + \hat{H}_{SO} + \hat{H}_Z \tag{1}$$

The zero field splitting of the lowest electronic states can be determined from the energies of the SOCI via mapping onto the eigenvalues of appropriate spin Hamiltonians.[4] The principal values of the g-tensor can be calculated by means of field-dependent energy differences[13], or by transformation of the Zeeman matrix into the basis of the lowest Kramers doublet.[7]

Modified CASCI Approach for the Calculation of Exchange Interactions

The magnetic exchange coupling originates from the coupling of the spins of different transition metal centers in bi- and trinuclear complexes and is calculated within the modified CASCI approach[15] schematically shown in Fig. 1.

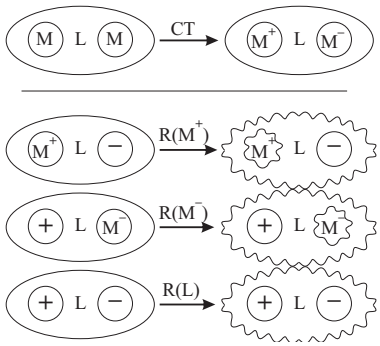


FIGURE 1. Sketch of the modified CASCI approach: M, M^+, and M^- represent the metal centers in their usual oxidation state and in the oxidation states they possess in the CT states, respectively. $+$ and $-$ symbolize large core pseudo potentials with the same charge.

In a standard CASSCF calculation the metal to metal charge transfer (MMCT) states, which are responsible for the superexchange contribution of the magnetic exchange coupling, are too high in energy. This is caused by missing orbital relaxation in the MMCT states. Besides some exceptional cases, it is not easily possible to perform calculations directly for a MMCT state. Therefore, the orbital relaxation contributions to the energy of the MMCT configurations are extrapolated by a set of ab initio calculations (line 2–4 in figure 1) on systems containing just one metal center or only ligands. The neglected metal centers are substituted by large core pseudo potentials. The energy of the MMCT states is corrected by adding the relaxation energies of one metal center and the ligands obtained in line 2 and 3 and substractiong the doubly counted relaxation energy of the ligands determined in line 4. The diagonal matrix elements of the MMCT determinats in the CI matrix are then shifted by the relaxation energy.

In the MCASCI calculations, the active space is constructed from the d-orbitals of each transition metal center. First, a state average CASSCF calculation on the high-spin states is performed. A Boys localization [18] of the active orbitals allows to address the MMCT determinats.

The procedure allows to calculate magnetic exchange coupling constants of the same quality as with the much more time consuming CASPT2 or MCCEPA methods. The resulting wave functions can be used in a subsequent SOCI calculation.

Simulation of Magnetic Data

Molar magnetic susceptibilities can be simulated from a field-dependent eigenspectrum by Boltzmann-averaging over numerical differentials:[8]

$$\chi_u(T) = -\frac{N_A}{B^2}\frac{\sum_n (E_n(B_u) - E_n(B=0)) e^{-E_n(B_u)(kT)^{-1}}}{\sum_n e^{-E_n(B_u)(kT)^{-1}}} \tag{2}$$

Eigenvalues of a Heisenberg-Dirac-van Vleck-Hamiltonian, as well as ab initio energies can be used to calculate susceptibilities. Thus, a direct comparison and validation of spin Hamiltonian parameters with theory and experiment is possible.

RESULTS AND DISCUSSION

In Tab. 1 zero-field splittings and g-values were summarized for a set of quasi-octahedrally and tetrahedrally coordinated Co and Ni containing mononuclear transition metal complexes. As also observed by Boča [14] the axial zero-field splitting parameter D covers a large variety of values.

TABLE 1. Axial zero-field splitting parameter obtained by spin-orbit configuration interaction calculations via mapping onto the spin Hamiltonian $\hat{H}_{\text{spin}} = D\hat{S}_z^2$ with $S = \frac{3}{2}$, and $S = 1$, respectively. The signs in parenthesis are not well-defined due to strong mixing of different M_S components.

Complex	**Reference**	$^1\Gamma$	$^2\Gamma$	$^3\Gamma$	D	**g**
$(CoCl_6)^{4-}$	[13]	0.0	372	–	−166	4.3
$(CoMg_2(SPh)_8)^{2-}$	[5]	0.0	63.24	–	(−)31.6	2.1
(CoMg(bipym*)(hfac)$_2$)	[6]	0.0	151.8	–	(+)75.9	4.5
$(VMg_2(SPh)_8)^{2-}$	[5]	0.0	0.11	–	+ 0.05	2.0
$(NiMg_2(SPh)_8)^{2-}$	[5]	0.0	31.59	34.10	+ 32.8	2.0
$(MgNi_2(SPh)_8)^{2-}$	[5]	0.0	32.86	36.21	+ 34.5	2.0

Furthermore, we investigated the magnetic exchange coupling constants of the full bi- and trinuclear complexes (Fig. 2). The magnetic data is visualized in Fig. 3. The direct simulation of μ_{eff}-characteristics and different effective spin Hamiltonian models allowed for a deeper understanding of the electronic coupling of the three metal centers in these complexes.[5] By comparison to a hypothetical symmetric compound (blue line in Fig. 3), we could show the influence of the asymmetric distortion of the Ni–V–Ni complex on the magnetic data.

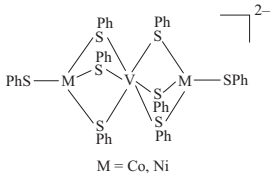


FIGURE 2. Schematic picture of the trinuclear complexes investigated.

CONCLUSIONS

We have presented a protocol to obtain magnetic data of polynuclear transition metal compounds by ab initio multireference methods. The procedure consists of structure determination by DFT, CASSCF and SOCI calculations on model complexes with only one active magnetic center for the determination of single-ion zero field splittings and g-tensors, and MCASCI calculations combined with SOCI for the determination of the magnetic exchange coupling constants applied to polynuclear complexes. In the last step the magnetic data is obtained either by Boltzmann averaging or with a HDvV spin hamiltonian. The method was successfully applied to different polynuclear complexes and the resulting magnetic data was compared to experiment.

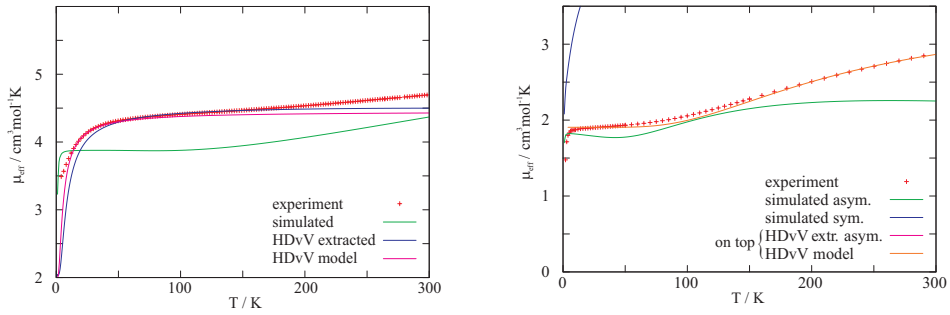


FIGURE 3. Effective magnetic moment of $(Co_2V(SPh)_8)^{2-}$ (left) and $(Ni_2V(SPh)_8)^{2-}$ (right). Shown are experimental as well as simulated data from a) ab initio energies, b) HDvV Hamiltonian $\hat{H} = -2\sum_{i<j} J_{ij}\hat{\mathbf{S}}_i\hat{\mathbf{S}}_j + \sum_i D_i\hat{S}^2_{z,i}$ with parameters extrated from the ab initio data ($(Co_2V(SPh)_8)^{2-}$: $J_{12} = J_{23} = -214.3$ cm^{-1}, and $(Ni_2V(SPh)_8)^{2-}$: $J_{12} = -681.5$ cm^{-1}, $J_{23} = -85.6$ cm^{-1}), and c) effective HDvV model (one-center for $(Co_2V(SPh)_8)^{2-}$, two-center with $J = -137.8$ cm^{-1} for $(Ni_2V(SPh)_8)^{2-}$).[5] For $(Ni_2V(SPh)_8)^{2-}$, the data for an symmetric and asymmetric structure (with respect to the Nickel-Vanadium distance) is shown for comparison.

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